Task 06: Minimal 1D Diffusion Model

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1 Introduction

The goal of this task was to implement a minimal diffusion model for one-dimensional data. The target distribution is a mixture of two Gaussian modes centered at -4 and 4 with unit standard deviation. This setting is intentionally simple, but it makes the main idea of diffusion models visible: a fixed forward process gradually adds noise, and a learned reverse process transforms noise back into samples from the target distribution.

2 Method

The forward diffusion process used a fixed variance schedule with

$$\beta_t = 0.02, \qquad T = 250.$$

For a clean sample x_0, noisy samples were generated directly using

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \qquad \epsilon \sim \mathcal{N}(0, 1),$$

where $\bar{\alpha}_t = \prod_{s=1}^{t} (1 - \beta_s)$.

The denoising model was a small fully connected MLP. It receives the noisy scalar value x_t and the normalized timestep t/T, and predicts the noise ϵ that was added. The model was trained with mean squared error loss and Adam. After training, samples were generated by starting from standard Gaussian noise and applying the learned reverse denoising process step by step. The notebook was run for two target mixtures: a balanced 50/50 mixture and an imbalanced 25/75 mixture. The implementation is available in the accompanying GitHub repository notebook.

3 Results

Figure 1 shows the fixed forward diffusion process for the imbalanced target distribution, where the left mode has probability 0.25 and the right mode probability 0.75. The two modes are clearly visible at early timesteps. As the timestep increases, Gaussian noise dominates and the original mixture structure gradually disappears. This confirms that the forward process performs the intended corruption from data to noise.

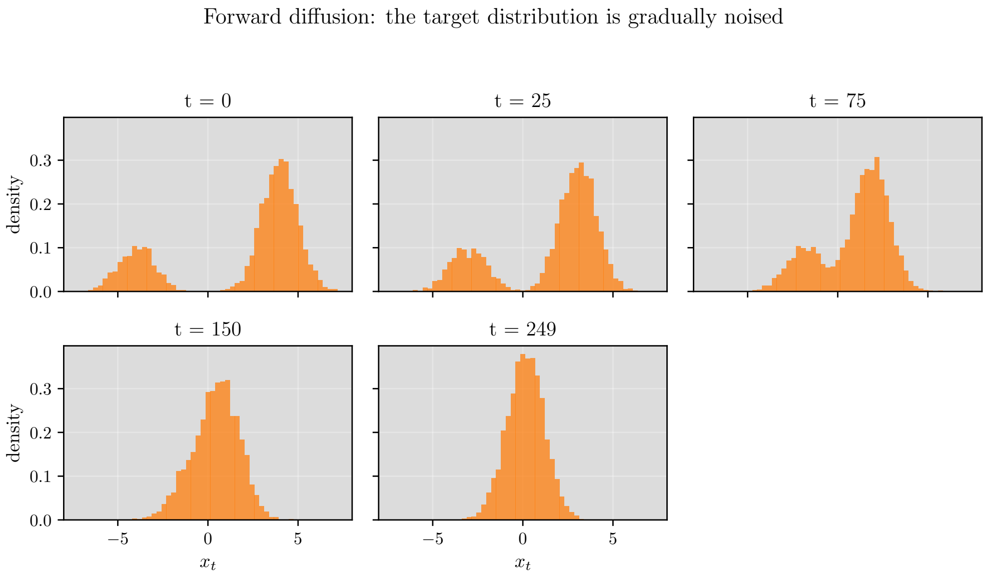


Figure 1: Forward diffusion snapshots for the imbalanced mixture. The original two-mode structure is gradually destroyed as the timestep increases.

Figure 2 shows the learned reverse diffusion process for the same imbalanced target distribution. The initial samples are close to standard Gaussian noise. During reverse diffusion, the distribution gradually separates into two modes, and the larger right mode is preserved.

Figure 3 compares the final generated distribution with the true target distribution for the two notebook runs. In the balanced case, the model recovers two modes with similar probability mass. In the imbalanced case, it also learns the unequal mode weights, showing that the same simple diffusion setup is not restricted to symmetric targets.

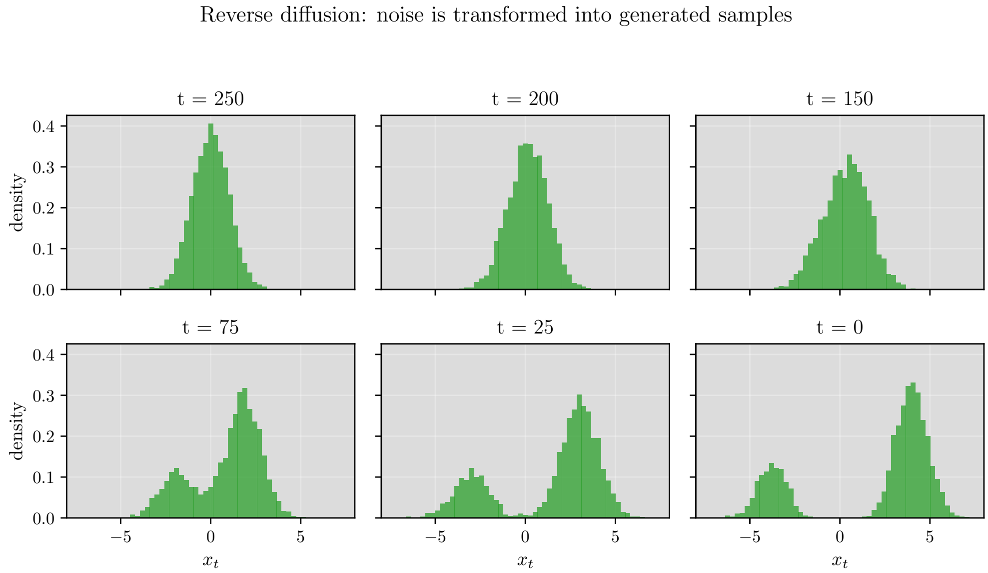
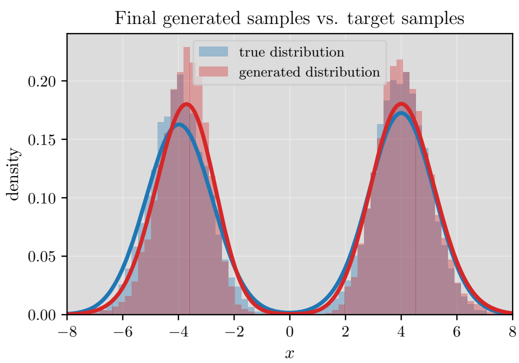
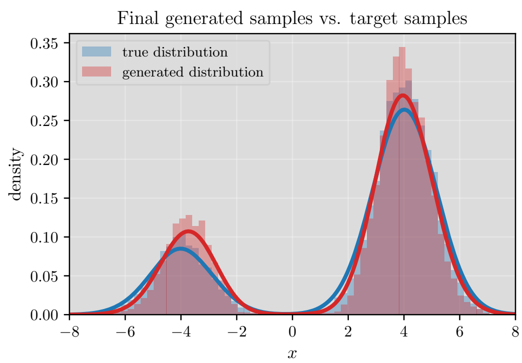


Figure 2: Reverse diffusion snapshots for the imbalanced mixture with 25% of samples in the left mode and 75% in the right mode.



(a) Balanced mixture, 50/50.



(b) Imbalanced mixture, 25/75.

Figure 3: Final generated samples compared with the true target distribution. The model captures the two-mode structure in both settings and also follows the unequal mode probabilities in the imbalanced case.

4 Conclusion

The implemented 1D diffusion model successfully learned to generate samples from a two-Gaussian target distribution. Although the model is very small and uses only scalar timestep conditioning, it captures both the balanced and imbalanced mixtures reasonably well. The main limitation is that the setup is intentionally minimal; possible improvements include richer timestep embeddings, a learned or scheduled β_t, longer training, and quantitative distribution metrics in addition to visual comparisons.